

MD. HASAN IMON

Aspiring AI/ML Engineer | AI/ML Developer

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SUMMARY

AI/ML Engineer specializing in Generative AI, and NLP with expertise in building production-ready multi-agent systems, RAG pipelines, and tool-augmented AI solutions. Skilled in fine-tuning LLMs (PEFT, LoRA, QLoRA) and applying NLP techniques with sentiment analysis, transformers, language modeling. Proven track record of creating enterprise-grade AI applications that decision-making, improve efficiency, and deliver measurable business outcomes. Passionate about developing scalable, real-world AI products that create impact.

EXPERIENCE

Research Assistant (AI/ML) | City University | June 2025 – September 2025

- Authored a research paper "Boosting Security: Gradient Boosting Models for Advanced Malware Detection". Malware detection system leveraging ensemble & boosting techniques to identify malware.
- Achieved state-of-the-art performance with XGBoost, 80% accuracy, F1-score 0.80 which outperforming classical ML models. Designed full pipeline: data preprocessing, EDA, feature selection, model training & evaluation using accuracy, precision, recall, and F1-score.

PROJECTS

MediGenius: Multi-Agent Medical Assistant – [GitHub](#)

Build an AI-driven medical assistant with Multi agent communication control.

- Achieving 82% clinical alignment, reducing misinformation by ~60% and improving patient query resolution rate by ~75%, driving higher trust, Context-aware, multi-turn consultations and adoption with enabling real-time empathetic medical support AI-driven healthcare consultation.
- Architected a modular, tool-augmented agentic ecosystem integrating Planner, Retriever, Answer Generator, and Fallback Agents with medical RAG over verified PDFs, SQLite with long term memory, and a Flask, HTML, CSS and JS deployed via Dockerized CI/CD pipeline.

InformaTruth: AI-Driven News Authenticity Analyzer – [GitHub](#)

Build explainable fake news detection system using Fine-tuned datasets.

- Multi-agent fact-checking system that automatically verifies news, PDFs, and web content, achieving ~70% accuracy and F1 ~69% on the LIAR dataset. Implemented with RoBERTa fine-tuning, LangGraph, RAG pipelines, fallback retrieval agents, FastAPI endpoints, Dockerized for deployment with Web UI.
- Boosted business impact and reliability by integrating memory, multi-agent tool routing, and fallback workflows, improving query coverage to 95% and answer reliability by ~60%, enabling enterprise-grade automated fact verification at scale.

AutoDocThinker: Intelligent Agentic Search Engine – [GitHub](#)

Build an intelligent document QA system using Agentic RAG

- Developed Q&A system that provides accurate answers from PDFs, DOCX, TXT, and URLs, achieving ~95% context retrieval coverage and reducing fallback calls to Wikipedia to ~5%, enabling faster and more reliable information access for enterprise users.
- Stateful multi-agent workflow with dynamic tool routing, supporting ingestion, retrieval, LLM answering, and fallback agents. Engineered with vector-based similarity search, recursive text splitting improving query processing efficiency by ~40% while handling 50+ simultaneous documents with Web apps.

TrueWealth AI: Your AI-Powered Financial Strategist – [GitHub](#)

Multi-agent financial advisor with Query breakdown & multi-stage planning.

- Built an AI-powered multi-agent financial advisor that delivers reliable, real-time investment insights by combining LLM reasoning, Agentic RAG and fallback web/news search. Achieving 99% query coverage with <3s P95 latency, reducing manual research effort for clients.
- Enhanced with LangGraph orchestration, HuggingFace embeddings, ChromaDB, Groq LLM, and modular agents (Planner, Retriever, Executor, Memory, Tool Router), deployed via Docker + FastAPI + Web UI, ensuring scalability, resilience, and enterprise readiness.

SKILLS

- ✓ **Programming & Data:** Python, Data Analytics, Pandas, NumPy, Matplotlib, Seaborn.
- ✓ **MLOps & Backend:** REST APIs, FastAPI, Flask, Git/GitHub, OOP, Docker, Postman
- ✓ **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Classification, Regression, Clustering, Model Evaluation, PyTorch, TensorFlow, Keras, Scikit-learn
- ✓ **NLP & Large Language Models (LLMs):** Text Classification, Embeddings, Hugging Face, NLTK, spaCy, HuggingFace Transformers, GPT, LLaMA, Mistral, Claude, Custom Fine-tuning (PEFT, LoRA, QLoRA)
- ✓ **AI Agents & Multi-Agent Systems:** LangChain, LangGraph, AgentOps, CrewAI, LlamaIndex, Agentic RAG Pipelines, Agent Orchestration, Tool-Augmented, Memory-Augmented, Prompt Engineering, HITL
- ✓ **Vector Databases & Storage:** FAISS, ChromaDB, Pinecone, Weaviate, MySQL, MongoDB
- ✓ **Cloud & Deployment:** Docker, CI/CD pipelines, Render

EDUCATION

Bachelor of Computer Science & Engineering

City University, Bangladesh (2022 - 2025)

CERTIFICATION

- **Machine Learning Specialization** by Stanford University (Coursera)
- **Programming for Everybody** by University of Michigan (Coursera)